

Civil Engineering

Calculus and Differential Methods

Learner Notes

Differentiation, integration, ODEs, and numerical methods for civil engineering.

Learning outcome	Focus
Perform calculus operations	Applied mathematics in workplace and training contexts.
Solve ordinary differential equations	Applied mathematics in workplace and training contexts.
Perform numerical methods	Applied mathematics in workplace and training contexts.

1. Topic Overview

This topic develops mathematical skills required in Civil Engineering. Learners should show clear method, correct units, and technical interpretation.

2. Key Formulas

$$\frac{d}{dx}(ax^n) = anx^{n-1}$$

$$\int ax^n dx = \frac{ax^{n+1}}{n+1} + C$$

$$\frac{dy}{dx} + P(x)y = Q(x)$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

3. Worked-Example Method

Step 1: identify the given data. Step 2: select the correct formula. Step 3: substitute values carefully. Step 4: compute. Step 5: interpret the answer with correct units.

$$\frac{d}{dx}(ax^n) = anx^{n-1}$$

4. Practice Questions

- Find the gradient function of a road profile.

2. Determine area under a drainage channel curve.
3. Apply Newton-Raphson method to find a root.